

PH 522: Statistical Mechanics

Course Code	PH 522
Course Name	Statistical Mechanics
L-T-P-C	3-0-0-3
Prerequisites	Undergraduate thermodynamics
Intended for	Core for M.Sc. and I-Ph.D. in Physics; Elective for others
Mutual Exclusion	None
Approval	9th Senate (May 2016), in the Minor in Physics curriculum
Revised	64th BoA (May 2026) - credits changed to 3-0-0-3

Preamble

Statistical mechanics use probability methods to extend the mechanics to many-body systems for statistical predictions of their collective behavior. This course introduces basic probability theory concepts and, with a thermodynamics review, thus acts as a bridge between thermodynamics and the mechanics of constituent particles. The topics covered include Statistical mechanics of ideal gas systems, harmonic oscillators, spin models, etc., providing the primary functioning of the formalisms of statistical mechanics for ideal systems. Methods of statistical mechanics serve as essential prerequisites to many advanced topics in physics, theoretical chemistry, and engineering that use probabilistic predictions on many-body problems. This course is at an advanced level with more mathematical proofs that enhance the basic introduction of thermodynamics and statistical mechanics at the undergraduate level.

Course Modules

- **Unit/Topic 1: Review of Thermodynamics:** Laws of Thermodynamics, Carnot cycle, entropy Specific heat, Maxwell relations, Thermodynamic potentials, Ideal gas, Equation of state, van der Waals equations, Gibbs-Duhem relation Thermodynamic stability (6 Hours)
- **Unit/Topic 2:** Probability concepts and examples - random variables single and multiple Probability distribution of many variables Binomial distribution and One Dimensional random walk problem - probability distributions and moments. Law of large numbers and Central Limit theorem. Information and entropy. (6 Hours)
- **Unit/Topic 3:** Liouville equation-Boltzman ergodic hypothesis, Gibbsian ensemble. Phase space and connection between mechanics and statistical mechanics by thermodynamic equilibrium. Microcanonical ensemble, microstates, and macrostates. Classical ideal gas, phase space volume of the ideal gas. Gibb's Paradox. (6 Hours)
- **Unit/Topic 4:** Canonical ensemble, partition function. Helmholtz free energy, Thermodynamics from the partition function. Classical ideal gas- equipartition and Virial theorem. Examples: harmonic oscillator and spin systems, Grand canonical ensemble-density, and energy fluctuations- Gibbs free energy. (10 hours)
- **Unit/Topic 5:** Formulation of quantum statistical mechanics density matrix- micro-canonical, canonical, and grand canonical ensembles- Maxwell-Boltzmann, Fermi-Dirac, and Bose-Einstein statistics - comparison. The density matrix of a two-particle ideal gas - comparison in different statistics. Ideal gas in classical and quantum ensembles. (6 Hours)
- **Unit/Topic 6:** Ideal Bose and Fermi systems Examples of ideal quantum gases. Bose-Einstein and Fermi-Dirac functions and approximations. Bose-Einstein condensation, statistics of phonons in ideal solids, Black body radiation, Fermi Energy. (8 Hours)

Laboratory/practical/tutorial Modules: None

Text Books

1. R K Pathria and Paul D Beale, *Statistical Mechanics*, Academic Press US, 2011.
2. M Kardar *Statistical Physics of Particles* Cambridge University Press, UK, 2007.

References

1. Kerson Huag, *Statistical Mechanics*, Wiley India, India, 2000.
2. *Fundamentals of Statistical and Thermal Physics*, F Reif, Waveland Press US, 2010.
3. *Statistical Physics part I*, L D. Landau and E. M. Lifshitz, Elsevier, India 2011.